

Lab Sheet 3: SQL Functions

A. Number Functions

1. ABS()

sql

```
SELECT ABS(-45.67) AS Absolute_Value_1,  
       ABS(120) AS Absolute_Value_2;
```

Use code with caution.

2. ROUND()

sql

```
SELECT ROUND(123.4567, 2) AS Round_Two_Decimals,  
       ROUND(123.75) AS Round_To_Integer;
```

Use code with caution.

3. CEIL() and FLOOR()

sql

```
SELECT CEIL(25.12) AS Ceil_Value,  
       FLOOR(25.99) AS Floor_Value;
```

Use code with caution.

4. MOD()

sql

```
SELECT MOD(17, 5) AS Remainder_1,  
       MOD(20, 2) AS Remainder_2;
```

Use code with caution.

5. Rounded Salary

sql

```
SELECT Emp_Name,  
       Salary,  
       ROUND(Salary, 0) AS Rounded_Salary
```

```
FROM Employee;
```

Use code with caution.

B. Aggregate Functions

6. COUNT()

sql

```
SELECT COUNT(Student_ID) AS Total_Students  
FROM Student;
```

Use code with caution.

7. SUM()

sql

```
SELECT SUM(Salary) AS Total_Salary_Expenditure  
FROM Employee;
```

Use code with caution.

8. AVG()

sql

```
SELECT AVG(Salary) AS Average_Salary
FROM Employee;
Use code with caution.
```

9. MAX()

sql

```
SELECT MAX(Salary) AS Highest_Salary
FROM Employee;
Use code with caution.
```

10. MIN()

sql

```
SELECT MIN(Salary) AS Lowest_Salary
FROM Employee;
Use code with caution.
```

C. Character Functions

11. UPPER()

sql

```
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM Student;
Use code with caution.
```

12. LOWER()

sql

```
SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM Student;
Use code with caution.
```

13. LENGTH()

sql

```
SELECT Student_Name,
       LENGTH(Student_Name) AS Character_Count
FROM Student;
Use code with caution.
```

14. CONCAT()

sql

```
SELECT CONCAT(First_Name, ' ', Last_Name) AS Full_Name
FROM Student_Details;
Use code with caution.
```

15. SUBSTRING() / SUBSTR()

sql

```
SELECT SUBSTRING('Database Management System', 1, 8) AS  
Substring_Sample;  
Use code with caution.
```

D. Conversion Functions

16. Number to Character

sql

```
SELECT CAST(1050.75 AS CHAR) AS Number_To_String;  
Use code with caution.
```

17. String to Numeric

sql

```
SELECT CAST('789.50' AS DECIMAL(10,2)) AS String_To_Decimal;  
Use code with caution.
```

18. Date Format Conversion

sql

```
SELECT DATE_FORMAT(NOW(), '%d-%m-%Y') AS Formatted_Date;  
Use code with caution.
```

19. CAST()

sql

```
SELECT CAST('2026-09-23' AS DATE) AS Text_To_Date,  
       CAST(55.88 AS INT) AS Decimal_To_Int;  
Use code with caution.
```

20. CONVERT()

sql

```
SELECT CONVERT('2026-09-23', DATE) AS Converted_Date;  
Use code with caution.
```

E. Date Functions

21. Current Date

sql

```
SELECT CURRENT_DATE() AS Today_Date;  
Use code with caution.
```

22. Extract Year, Month, Day

sql

```
SELECT EXTRACT(YEAR FROM CURRENT_DATE) AS Current_Year,  
       EXTRACT(MONTH FROM CURRENT_DATE) AS Current_Month,  
       EXTRACT(DAY FROM CURRENT_DATE) AS Current_Day;
```

Use code with caution.

23. Date Difference

sql

```
SELECT DATEDIFF('2026-12-25', '2026-09-23') AS Days_Until_Christmas;
```

Use code with caution.

24. Add Days to Date

sql

```
SELECT DATE_ADD('2026-09-23', INTERVAL 10 DAY) AS New_Date;
```

Use code with caution.

25. Format Employee Joining Date

sql

```
SELECT Emp_Name,  
       DATE_FORMAT(Joining_Date, '%d/%m/%Y') AS Formatted_Joining_Date  
FROM Employee;
```

OUTPUT:-

A. Number Functions

1. ABS()

Absolute_Value_1	Absolute_Value_2
------------------	------------------

45.67	120
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2. ROUND()

Round_Two_Decimals	Round_To_Integer
--------------------	------------------

123.46	124
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3. CEIL() and FLOOR()

Ceil_Value	Floor_Value
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26	25
----	----

4. MOD()

Remainder_1	Remainder_2
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2	0
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5. Rounded Salary

(Assuming Alice Smith's salary became 52800.00 and Charlie Brown's became 66000.00 after the previous 10% raise operation)

Emp_Name	Salary	Rounded_Salary
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Alice Smith	52800.00	52800
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Charlie Brown	66000.00	66000
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B. Aggregate Functions

6. COUNT()

(Based on the remaining active student records from Lab 1)

Total_Students

7

7. SUM()

Total_Salary_Expenditure

118800.00

8. AVG()

Average_Salary

59400.00

9. MAX()

Highest_Salary

66000.00

10. MIN()

Lowest_Salary

52800.00

C. Character Functions

11. UPPER()

Student_Name Uppercase_Name

Amit Sharma AMIT SHARMA

Rahul Verma RAHUL VERMA

12. LOWER()

Student_Name Lowercase_Name

Amit Sharma amit sharma

Rahul Verma rahul verma

13. LENGTH()

Student_Name Character_Count

Amit Sharma 11

Rahul Verma 11

14. CONCAT()

(Assuming a generic row where First_Name = 'John' and Last_Name = 'Doe')

Full_Name

John Doe

15. SUBSTRING() / SUBSTR()

Substring_Sample

Database

D. Conversion Functions

16. Number to Character

Number_To_String

"1050.75"

17. String to Numeric

String_To_Decimal

789.50

18. Date Format Conversion

Formatted_Date

23-09-2026

19. CAST()

Text_To_Date Decimal_To_Int

2026-09-23 55

20. CONVERT()

Converted_Date

2026-09-23

E. Date Functions

21. Current Date

Today_Date

2026-09-23

22. Extract Year, Month, Day

Current_Year Current_Month Current_Day

2026 9 23

23. Date Difference

Days_Until_Christmas

93

24. Add Days to Date

New_Date

2026-10-03

25. Format Employee Joining Date

(Assuming a sample date of 2025-01-15 stored in the system)

Emp_Name Formatted_Joining_Date

Alice Smith 15/01/2025

Charlie Brown 20/03/2024